SDS Exercise Sheet 6:
LLMs

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1 Introduction

In this exercise you are going to conduct task-oriented dialogue with LLMs and draw comparisons to modular dialogue systems.

The objective of this practical are as follows:

- Learn how LLMs, e.g. ChatGPT, can be used to conduct task-oriented-style dialogues.
- Identify the key differences between instruction-tuned LLMs and modular dialogue systems.
- Assess the consequences of the differences above and how it should be considered when deploying dialogue systems.

2 Exercises

Task 1: Analyzing Task-oriented dialogue sample

Analyze the following dialogue sample between a user and a modular task-oriented dialogue.

User goal: You are looking for a place to stay in **Cambridge**. The hotel should be in the **cheap** price range. The hotel should include **free parking**. Once you find the hotel you want to book it for **6 people** and **3 nights** starting from **Tuesday**. Make sure you get the **reference number**.

Utterance	Dialogue acts and API calls	Dialogue state
I am looking for a place to to stay that has cheap price range it should be in a type of hotel	**inform**(price range=cheap) **inform**(type=hotel)	price range: cheap type: hotel
Okay, do you have a specific area you want to stay in?	**request**(area)	price range: cheap type: hotel
no, i just need to make sure it's cheap. oh, and i need parking	**inform**(parking=yes)	price range: cheap type: hotel parking: yes
I found 1 cheap hotel for you that includes parking. It is Holiday Inn. Do you like me to book it?	**databasequery** **inform**(price range=cheap) **inform**(choice=1) **inform**(parking) **inform**(name=holiday inn) **inform**(type=hotel)	price range: cheap type: hotel parking: yes
Yes, please. 6 people 3 nights starting on tuesday.	**inform**(book people=6) **inform**(book stay=3) **inform**(book day=tuesday)	price range: cheap type: hotel parking: yes book people: 6 book stay: 3 book day: tuesday

Booking was successful. Reference number is : 00000028. Anything else I can do for you?	**book** **inform**(ref=00000028) **reqmore**	price range: cheap type: hotel parking: yes book people: 6 book stay: 3 book day: tuesday ref: 00000028
No, that will be all. Good bye.	**bye**	price range: cheap type: hotel parking: yes book people: 6 book stay: 2 book day: tuesday ref: 00000028
Thank you for using our services.	**bye**	price range: cheap type: hotel parking: yes book people: 6 book stay: 2 book day: tuesday ref: 00000028

Task 2: Freeform interaction with web-enabled LLMs

Evaluate the behavior of an open-domain LLM with live web access when tasked with a specific, closed-domain booking goal. Interact with an instruction-tuned LLM to achieve the same user goal from Task 1. For example, you may use the free version of ChatGPT or Gemini. In addition to a replicating the interaction above, actively test its boundaries, or lack thereof, using various dialogue probes. For example, you can try to find out how it safeguards against malicious user, how it guarantees any bookings made, or how it ranks competing hotel providers. Try a **minimum of 2 probes**.

Task 3: Structured prompting with LLMs

Now let's emulate the modules of a task-oriented dialogue system pipeline with LLMs. Interact to fulfill the same user goal. Prompt the LLM to produce the intermediate representations before producing the final response.

A sample prompt is provided below. Refine it to incorporate the concepts you have learned from the lectures so far, such that the LLM can perform goal completion as in the dialogue in Task 1. You can find the database in json format in sciebo (`hotel_db_small.json`), to be included as part of the prompt. Continue the dialogue over multiple turns in order to fulfill the user goal.
Sample prompt:

> You are a modular dialogue system for hotel booking. You will respond in 4 steps:
> NLU: Identify intent and extract slot values (location, price range, party size, time) from the user's input.
> DST: Update the dialogue state with current slot values.
> Policy: Decide the next system action (e.g., ask for missing slots, confirm info, or call API).
> NLG: Generate the natural language response.
> Here is the database of hotels: [INSERT DATABASE HERE]
> Here is your current dialogue state: [INSERT EMPTY DIALOGUE STATE HERE]
> User: [INSERT FIRST TURN HERE].

3 Discussion

In your final report, write a short report covering, but not limited to, the following points. Additional insights, analysis, and discussion are highly encouraged.

- Compare the dialogues from Tasks 1, 2, and 3.
  - What are the main differences in terms of naturalness, flexibility, and controllability? You may provide dialogue snippets to demonstrate your points.
  - Is the user/are you able to fulfill the goal? Explain.
  - Is it sufficient to replace modular task-oriented systems with structured prompting on LLMs? Explain.
- Discuss the probes you used to complete Task 2 and your observations of the outcome. How does this change in Task 3?
- Provide the final prompt you use to complete Task 3, and explain the reasoning behind it.
- Based on your observations in Task 2 and Task 3, are there risks when deploying a web-enabled LLM as a front-end customer service agent? Explain.